Unit 5

Knowledge Representation and Reasoning for various AI applications

Prepared by

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Agenda

- *Statistical Reasoning:*
  - ***Probability and Bayes' Theorem***
  - ***Certainty Factors and Rule-Base Systems***
  - ***Bayesian Networks***
  - ***Dempster-Shafer Theory***
- *Fuzzy Logic*
  - ***AI for knowledge representation***
  - ***Rule-based knowledge representation, procedural and declarative knowledge***
  - ***Logic programming, Forward and backward reasoning.***
- *Applications: Expert Systems, Decision Support System, Generative AI in Natural Language Processing (NLP) AI-based Programming tools.*

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Statistical reasoning

- Statistical reasoning involves the use of statistical methods to analyze and interpret data, draw conclusions, and make decisions under uncertainty.
- It helps AI systems and data-driven models handle uncertainty and variability in the real world.

Concept	Description	Example
Data	Collection of observations or measurements	Exam scores, sensor readings
Population	Entire set of items being studied	All students in a school
Sample	Subset of population used for analysis	50 randomly chosen students
Parameter	Numerical value describing a population	True average score
Statistic	Value calculated from a sample	Sample mean = 75

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Types of Statistical Reasoning

Descriptive Statistics

- Summarizes data using:
  - **Measures of central tendency:** Mean, Median, Mode
  - **Measures of dispersion:** Variance, Standard Deviation, Range
  - **Data visualization:** Bar charts, Histograms, Box plots
- **Purpose:** Describe and summarize information.

Inferential Statistics

- Draws conclusions about a population from sample data.
- Uses **probability theory** to estimate, predict, or test hypotheses.

Examples:

- Estimating average income of a city using a small sample.
- Testing if a new drug is more effective than an existing one.

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Probability in Statistical Reasoning

- Probability provides a **mathematical framework** to quantify uncertainty.
- **Experiment:** Any process that generates outcomes (e.g., coin toss)
- **Sample Space (S):** All possible outcomes (e.g., {H, T})
- **Event:** Subset of outcomes (e.g., getting a head)
- **Probability:** $P(A) = \frac{Favorable\ outcomes}{Total\ outcomes}$

Conditional Probability

- Represents how the probability of one event depends on another.
- $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$
- **Example:** Probability of being sick given that a test is positive.

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Bayes' Theorem in Artificial Intelligence

- Bayes' theorem is a fundamental concept in probability and statistics.
- It often referred to as Bayes' rule or Bayes' law, Bayesian reasoning, which determines the probability of an event with uncertain knowledge.
- Bayes' theorem is a mathematical formula for calculating conditional probability.
- It updates an initial probability ("prior") with new evidence to get a revised probability ("posterior").
- The general formula for Bayes' theorem is: $$\boxed{P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}}$$

- $P(A|B)$ is the **posterior probability** (probability of A given B).
- $P(A)$ is the **prior probability** (initial probability of A).
- $P(B|A)$ is the **likelihood** (probability of B given A).
- $P(B)$ is the **marginal probability** (total probability of B)

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Suppose you have a bag containing the following marbles:3 red marbles ,2 blue marbles,5 green marbles

In total, there are 3+2+5=10 marbles.

Step 1: Define Events

Let A be the event "drawing a red marble."

Let B be the event "drawing a marble that is not blue."

Step 2: Determine Probabilities

- **P(A): Probability of drawing a red marble**
  There are 3 red marbles out of 10 total. P(A)=3/10
- **P(B): Probability of drawing a marble that is not blue**
  Marbles that are not blue are red or green (3 red + 5 green = 8 marbles).
  P(B)=8/10
- **P(B|A): Probability of drawing a marble that is not blue, given it is red**
  If the marble is red, it is automatically not blue. So, this probability is 1.
  $P(B|A)=1$
- **$P(A \cap B)$: Probability of drawing a red marble AND it is not blue**
  The only red marbles are also not blue, so this is the same as P(A) P(A∩B)=3/10

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Bayes' theorem states:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

$$P(A|B) = \frac{1 \times \frac{3}{10}}{\frac{8}{10}} = \frac{3/10}{8/10} = \frac{3}{8}$$

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From a standard deck of playing cards, a single card is drawn. The probability that the card is king is 4/52, then calculate posterior probability P(King|Face), which means the drawn face card is a king card.

A single card is drawn from a **standard deck (52 cards)**.

Probability that a card is a **King**, P(King)=4/52.

To find the **posterior probability** P(King|Face): Probability that the card is a **King**, given that it is a **Face card**.

Step 1: Define Events

Let K = event that the card is a **King**
F = event that the card is a **Face card** (Jack, Queen, or King)

Step 2: Find the Required Probabilities

Total cards = 52

Face cards = 3 per suit × 4 suits = **12** P(F)=12/52

Kings = 4 P(K)=4/52

Every King is a Face card, so P(F|K) = 1

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Step 3: Apply Bayes' Theorem

$P(K|F)=(P(F|K)\cdot P(K))/P(F)$

Substitute values:

$P(K|F)=((1)\times 4/52)/(12/52)$

Step 4: Simplify

$P(K|F)=(4/52)/(12/52)=4/12=1/3$

$P(King|Face)=1/3$

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Uses of Bayes' Rule in Artificial Intelligence

- Naive Bayes Classification-Widely used in natural language processing and text classification.
- Spam Email Filtering-Determines whether incoming emails are spam.
- Anomaly Detection-Identifies anomalies or outliers in datasets.
- Personalization-Updates user preferences in recommendation systems.
- Medical Diagnosis-Updates the likelihood of various diagnoses based on patient symptoms, test results, and medical history

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Certainty Factors (CF) in Expert Systems

- Certainty Factors are **numerical values** used in expert systems to express the **degree of confidence or certainty** in the truth or falsity of a statement or hypothesis.
- **Purpose:**
  They help represent and manipulate uncertain or incomplete information in a reasoning process.
- **Range of Values:**
  - **+1.0** → Definitely True
  - **0.0** → Completely Uncertain (no knowledge)
  - **–1.0** → Definitely False
- **Example Usage:**
  - CF = +0.8 → The expert is 80% confident the conclusion is true.
  - CF = –0.6 → The expert is 60% confident the conclusion is false.
- **Formula:**
  When multiple rules contribute to the same conclusion:

$CF_{combined} = CF_1 + CF_2 \times (1 - CF_1)$(used when both CFs are positive)

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Components of Certainty Factors

- **Belief (MB – Measure of Belief):**
  The degree of confidence that a proposition is **true**.
  - Range: 0 to +1
  - Example: MB = 0.8 means 80% confidence that the statement is true.
- **Disbelief (MD – Measure of Disbelief):**
  The degree of confidence that a proposition is **false**.
  - Range: 0 to +1
  - Example: MD = 0.6 means 60% confidence that the statement is false.
- **Uncertainty:**
  The lack of confidence in either belief or disbelief.
  - Calculated as:
- Uncertainty $= 1 - (MB + MD)$

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Certainty Factor Calculation

- Certainty Factors can be combined from **multiple sources of evidence.**
- When two or more rules lead to the same conclusion, their certainty factors can be **merged** to produce an overall certainty factor for that conclusion.

General Formula: $CF = MB - MD$

where,

- MB: Measure of Belief
- MD: Measure of Disbelief

Combining Certainty Factors

- When multiple rules or pieces of evidence support the same hypothesis, the certainty factors are **combined** as follows:
- **Case 1: Both CF_1 and CF_2 are positive (support the hypothesis)**

$$CF_{combined} = CF_1 + CF_2 \times (1 - CF_1)$$

- **Case 2: Both CF_1 and CF_2 are negative (refute the hypothesis)**

$$CF_{combined} = CF_1 + CF_2 \times (1 + CF_1)$$

- **Case 3: CF_1 and CF_2 have opposite signs (conflicting evidence)**

$$\boxed{CF_{combined} = \frac{CF_1 + CF_2}{1 - \min(|CF_1|, |CF_2|)}}$$

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An AI system determines whether a patient has *pneumonia*.

- **Rule 1:** If the patient has a cough,

$$CF_1 = +0.7$$

- **Rule 2:** If the patient has chest pain,

$$CF_2 = +0.6$$

- **Formula:** $CF_{combined} = CF_1 + CF_2 \times (1 -| CF_1 |)$
- **Calculation:** $CF_{combined} = 0.7 + 0.6 \times (1 - 0.7) = 0.7 + 0.6 \times 0.3 = 0.7 + 0.18 = 0.88$

The system is **88% certain** that the patient has pneumonia.

An expert system predicts if it will rain today.

- **Rule 1:** If humidity > 80% → $CF_1 = 0.7$
- **Rule 2:** If dark clouds are present → $CF_2 = 0.5$
- $CF_{combined} = 0.7 + 0.5 \times (1 - 0.7) = 0.7 + 0.15 = 0.85$

Result: There is an **85% chance** of rain.

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Rule-based system in AI

- A Rule-Based System (RBS) is one of the earliest and most widely used approaches in Artificial Intelligence for representing knowledge and performing reasoning.
- It works on a collection of “IF–THEN” rules to derive conclusions or make decisions — similar to how human experts reason logically.
- Rule-Based System is an AI system that uses a set of logical rules to represent knowledge and an inference engine to apply those rules to known facts in order to derive new information.

Structure of a Rule-Based System

A typical rule-based system has three main components:

(a) Knowledge Base

Contains facts and rules.

Facts represent known information.

Rules represent expert knowledge in IF–THEN format.

Example:

IF temperature > 100°C THEN alarm = ON

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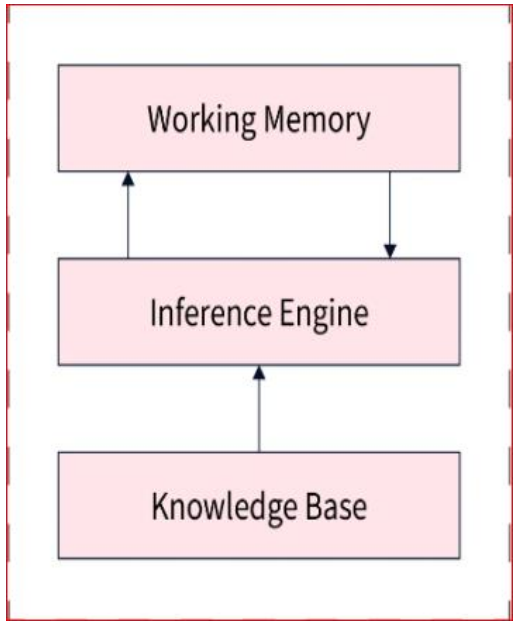
(b) Inference Engine

- The brain of the system.
- Applies rules to the known facts to derive new facts.
- Uses reasoning strategies such as:
  Forward chaining (data-driven)
  Backward chaining (goal-driven)



(c) Working Memory

- Temporary storage for facts that are currently known or derived during reasoning.
- Continuously updated as rules are applied.

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Reasoning Type	Description	Example
Forward Chaining	Start with known facts → apply rules → derive new facts → reach conclusion.	Data-driven approach used in **diagnostic systems**.
Backward Chaining	Start with goal/hypothesis → work backward using rules → verify if data supports it.	Goal-driven approach used in **Prolog** and **expert systems**.

Example

Rules:

1.IF it rains THEN road is wet.

2.IF road is wet THEN car slips.

Fact: It rains.

Using forward chaining → Conclusion: Car slips.

Functionality: The system checks the facts in the working memory against the rules in the knowledge base. When a rule's conditions are met, the inference engine executes the corresponding action or conclusion. These systems are commonly used in expert systems, decision support systems, and various AI applications.

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Applications of Rule-Based Systems

- **Expert Systems**

Expert systems are designed to emulate the decision-making abilities of human experts. They use a large set of rules to make inferences or recommendations.

Applications include:

  - **Medical Diagnosis:** Systems like MYCIN, an early expert system, assist in diagnosing diseases and recommending treatments based on patient data and medical knowledge.
  - **Financial Services:** Expert systems can assess credit risks, detect fraud, and provide investment advice.

- **Decision Support Systems**
  - These systems aid decision-making processes by providing relevant information and recommendations based on predefined rules. Examples include:
  - **Customer Support:** Automated customer support systems use rules to handle common queries and problems, providing quick and consistent responses.
  - **Manufacturing:** Rule-based systems help in managing production schedules, inventory control, and quality assurance.

- **Control Systems**

In control systems, rules govern the operation of machinery and equipment. Examples include:

  - **Automated Traffic Lights:** Rules determine the timing and sequencing of traffic lights based on traffic flow and other factors.
  - **Home Automation:** Smart home systems use rules to control lighting, heating, and security systems based on user preferences and sensor data.

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RULE BASED SYSTEMS	LEARNING-BASED SYSTEMS
Uses a set of predetermined rules to make decisions	Learns from data to make decisions
Rules are created by human experts	Knowledge is acquired through machine learning algorithms
Limited ability to adapt to new situations	Can adapt to new situations by learning from data
Transparent decision-making process	Decision-making process may not be transparent
Scalable by adding or modifying rules	Scalable by retraining the model
Consistent in decision-making	Decision-making may not always be consistent
Fast decision-making	May require significant time for training the model

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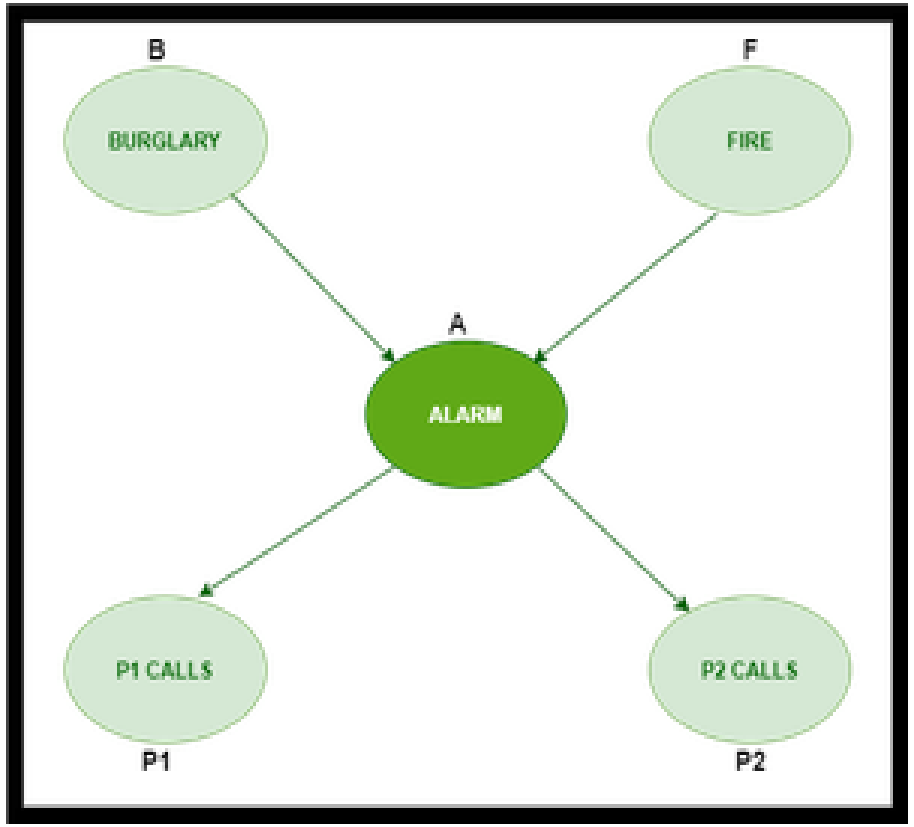

- An alarm 'A' - a node, say installed in a house of a person 'gfg', which rings upon two probabilities i.e burglary 'B' and fire 'F', which are - parent nodes of the alarm node.
- The alarm is the parent node of two probabilities P1 calls 'P1' & P2 calls 'P2' person nodes.
- Upon the instance of burglary and fire, 'P1' and 'P2' call person 'gfg', respectively. But, there are few drawbacks in this case, as sometimes 'P1' may forget to call the person 'gfg', even after hearing the alarm, as he has a tendency to forget things, quick.
- Similarly, 'P2', sometimes fails to call the person 'gfg', as he is only able to hear the alarm, from a certain distance.

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Burglary 'B' -

- **P (B=T) = 0.001** ('B' is true i.e burglary has occurred)
- **P (B=F) = 0.999** ('B' is false i.e burglary has not occurred)

Fire 'F' –

P (F=T) = 0.002 ('F' is true i.e fire has occurred)

P (F=F) = 0.998 ('F' is false i.e fire has not occurred)

B	**F**	**P (A=T)**	**P (A=F)**
T	T	0.95	0.05
T	F	0.94	0.06
F	T	0.29	0.71
F	F	0.001	**<u>0.999</u>**

Alarm 'A' -

Person 'P1' -

A	P (P1=T)	P (P1=F)
T	<u>0.95</u>	0.05
F	0.05	0.95

The person 'P1' node can be 'true' or 'false' (i.e may have called the person 'gfg' or not) . It has a parent node, the alarm 'A', which can be 'true' or 'false' (i.e may have rung or may not have rung ,upon burglary 'B' or fire 'F').

Person 'P2' -

A	**P (P2=T)**	**P (P2=F)**
T	<u>0.80</u>	0.20
F	0.01	0.99

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- The person 'P2' node can be 'true' or false' (i.e may have called the person 'gfg' or not). It has a parent node, the alarm 'A', which can be 'true' or 'false' (i.e may have rung or may not have rung, upon burglary 'B' or fire 'F').

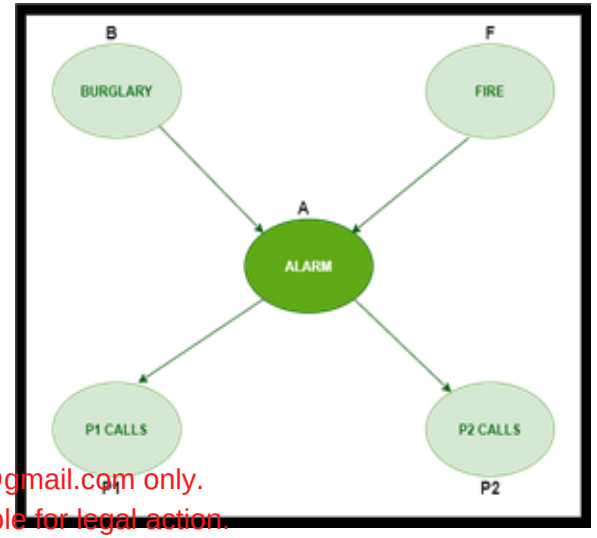
Find the probability that 'P1' is true (P1 has called 'gfg'), 'P2' is true (P2 has called 'gfg') when the alarm 'A' rang, but no burglary 'B' and fire 'F' has occurred.

=> **P (P1, P2, A, ~B, ~F)** [where- P1, P2 & A are 'true' events and '~B' & '~F' are 'false' events]

With respect to the question — **P (P1, P2, A, ~B, ~F)** , we need to get the probability of 'P1'. We find it with regard to its parent node - alarm 'A'. To get the probability of 'P2', we find it with regard to its parent node — alarm 'A'. We find the probability of alarm 'A' node with regard to '~B' & '~F' since burglary 'B' and fire 'F' are parent nodes of alarm 'A'.

P (P1, P2, A, ~B, ~F)

= P (P1/A) * P (P2/A) * P (A/~B~F) * P (~B) * P (~F)



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Dempster–Shafer Theory

- **Dempster–Shafer Theory (DST)** — also called the **Theory of Evidence** is a **mathematical framework for reasoning with uncertainty**.
- It was developed by **Arthur P. Dempster (1967)** and later extended by **Glenn Shafer (1976)**.
- It's an alternative to **Bayesian probability**, but instead of requiring precise probabilities for all events, DST allows **beliefs to be assigned to sets of possibilities**.
- DST deals with degrees of belief rather than exact probabilities.
- You can assign your belief mass to:
  - a single hypothesis (e.g., "the patient has flu"),
  - or to a set of hypotheses (e.g., "the patient has either flu or cold"), reflecting uncertainty or incomplete information.

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Terminologies and concepts

Concept	Description
Frame of Discernment (Θ)	The set of all possible hypotheses. Example: Θ = {Flu, Cold, Allergy}
Basic Probability Assignment (BPA) or **Mass function (m)**	Assigns a belief mass to each subset of Θ. Example: m({Flu}) = 0.4, m({Cold}) = 0.3, m({Flu, Cold}) = 0.3
Belief (Bel)	Degree of belief **that supports** a hypothesis. Calculated by summing all masses of subsets that are entirely within it.
Plausibility (Pl)	Degree to which evidence **does not refute** a hypothesis. It's the upper bound of belief.
Relationship	$(Pl(A) = 1 - Bel(\neg A)$

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Combining Evidence — **Dempster's Rule of Combination**

Let $\Theta = \{H_1, H_2, \ldots, Hn\}$

(a) Basic Probability Assignment (BPA)

$$\sum_{A \subseteq \Theta} m(A) = 1$$

and $m(\emptyset) = 0$

(b) Belief Function

$$Bel(A) = \sum_{B \subseteq A} m(B)$$

(c) Plausibility Function

$$Pl(A) = \sum_{B \cap A \neq \emptyset} m(B)$$

(d) Relationship

$$Pl(A) = 1 - Bel(\neg A)$$

When two independent sources of evidence m_1 and m_2 are available:

$$m(A) = \frac{1}{1-K} \sum_{B \cap C = A} m_1(B) \times m_2(C)$$

where

$$K = \sum_{B \cap C = \emptyset} m_1(B) \times m_2(C)$$

is the **conflict** between the sources.

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Example

A medical expert system is designed to diagnose whether a patient has Flu (F) or Allergy (A). Two tests provide uncertain evidence.

Test 1 (m_1):

$m_1(\{Flu\}) = 0.6$ $m_1(\{Allergy\}) = 0.2$ $m_1(\{Flu, Allergy\}) = 0.2$

Test 2

(m_2): $m_2(\{Flu\}) = 0.5$ $m_2(\{Allergy\}) = 0.3$ $m_2(\{Flu, Allergy\}) = 0.2$

Question: Combine both pieces of evidence using Dempster's rule of combination.

What is the final belief mass for {Flu}?

Based on the result, what diagnosis should the system prefer?

Frame of Discernment:
$\Theta = \{Flu\ (F), Allergy\ (A)\}$

Evidence	m({F})	m({A})	m({F,A})
Test 1 (m_1)	0.6	0.2	0.2
Test 2 (m_2)	0.5	0.3	0.2

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Medical Diagnosis

Compute all intersections and their product masses

B from m_1	C from m_2	Intersection	Product	Notes
{F}	{F}	{F}	0.6×0.5 = 0.30	—
{F}	{A}	∅	0.6×0.3 = 0.18	conflict
{F}	{F,A}	{F}	0.6×0.2 = 0.12	—
{A}	{F}	∅	0.2×0.5 = 0.10	conflict
{A}	{A}	{A}	0.2×0.3 = 0.06	—
{A}	{F,A}	{A}	0.2×0.2 = 0.04	—
{F,A}	{F}	{F}	0.2×0.5 = 0.10	—
{F,A}	{A}	{A}	0.2×0.3 = 0.06	—
{F,A}	{F,A}	{F,A}	0.2×0.2 = 0.04	—

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Compute the conflict factor (K)

$$K = Sum\ of\ products\ with\ empty\ intersection$$
$$K = 0.18 + 0.10 = 0.28$$
$$1 - K = 0.72$$

Compute the combined masses (normalize by 1−K)

$$m(F) = (0.30 + 0.12 + 0.10)/0.72 = 0.52/0.72 = 0.722$$
$$m(A) = (0.06 + 0.04 + 0.06)/0.72 = 0.16/0.72 = 0.222$$
$$m(F, A) = 0.04/0.72 = 0.056$$

Hypothesis	Combined Mass
m({Flu})	**0.72**
m({Allergy})	**0.22**
m({Flu, Allergy})	**0.06**

Interpretation

- **Belief in Flu (Bel(F)) = 0.72**
- **Belief in Allergy (Bel(A)) = 0.22**

Since the **belief in Flu** is **much higher**, the system should **diagnose the patient as likely having Flu**.

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Weather Forecasting

- Frame of discernment:
  Θ = {Rain (R), No Rain (N)}

Source	m({R})	m({N})	m({R, N})
Sensor 1	0.7	0.1	0.2
Sensor 2	0.6	0.3	0.1

B (m_1)	C (m_2)	Intersection	Product	Remark
{R}	{R}	{R}	0.7×0.6 = 0.42	—
{R}	{N}	Ø	0.7×0.3 = 0.21	Conflict
{R}	{R,N}	{R}	0.7×0.1 = 0.07	—
{N}	{R}	Ø	0.1×0.6 = 0.06	Conflict
{N}	{N}	{N}	0.1×0.3 = 0.03	—
{N}	{R,N}	{N}	0.1×0.1 = 0.01	—
{R,N}	{R}	{R}	0.2×0.6 = 0.12	—
{R,N}	{N}	{N}	0.2×0.3 = 0.06	—
{R,N}	{R,N}	{R,N}	0.2×0.1 = 0.02	—

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Compute total conflict K

$$K = 0.21 + 0.06 = 0.27$$

$$1 - K = 0.73$$

Combine masses (normalize)

$$m(R) = (0.42 + 0.07 + 0.12)/0.73 = 0.61$$

$$m(N) = (0.03 + 0.01 + 0.06)/0.73 = 0.14$$

$$m(R, N) = 0.02/0.73 = 0.03$$

Compute Belief and Plausibility

$$Bel(R) = m(R) = 0.61$$

$$Pl(R) = m(R) + m(R, N) = 0.61 + 0.03 = 0.64$$

Interpretation

- Belief that it will **Rain**: 61%
- Plausibility (possibility) that it could **Rain**: 64%

  → The system **should forecast rain** because belief is significantly higher than for "No Rain".

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Characteristics of Dempster Shafer Theory

- Ability to Handle Incomplete Information
- Independence from Prior Probability Assumptions
- Flexibility in Combining Evidence from Multiple Sources

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Fuzzy Logic

- Fuzzy Logic is an extension of classical (binary) logic that allows **partial truth values** between **0 and 1** — instead of being strictly *true (1)* or *false (0)*.
- It is used in AI to handle **uncertainty**, **approximation**, and **vagueness** in human reasoning just like how people think in real life.

In classical logic:

- "If temperature > 30°C, then it's hot."
- This rule fails when the temperature is **29.9°C** — it's *almost* hot, but not exactly.
- In **fuzzy logic**,
- Temperature = "Hot" is **0.8 true** (partially hot).
  Temperature = "Warm" is **0.6 true** (also partially warm).
- This allows systems to make smoother, more human-like decisions.

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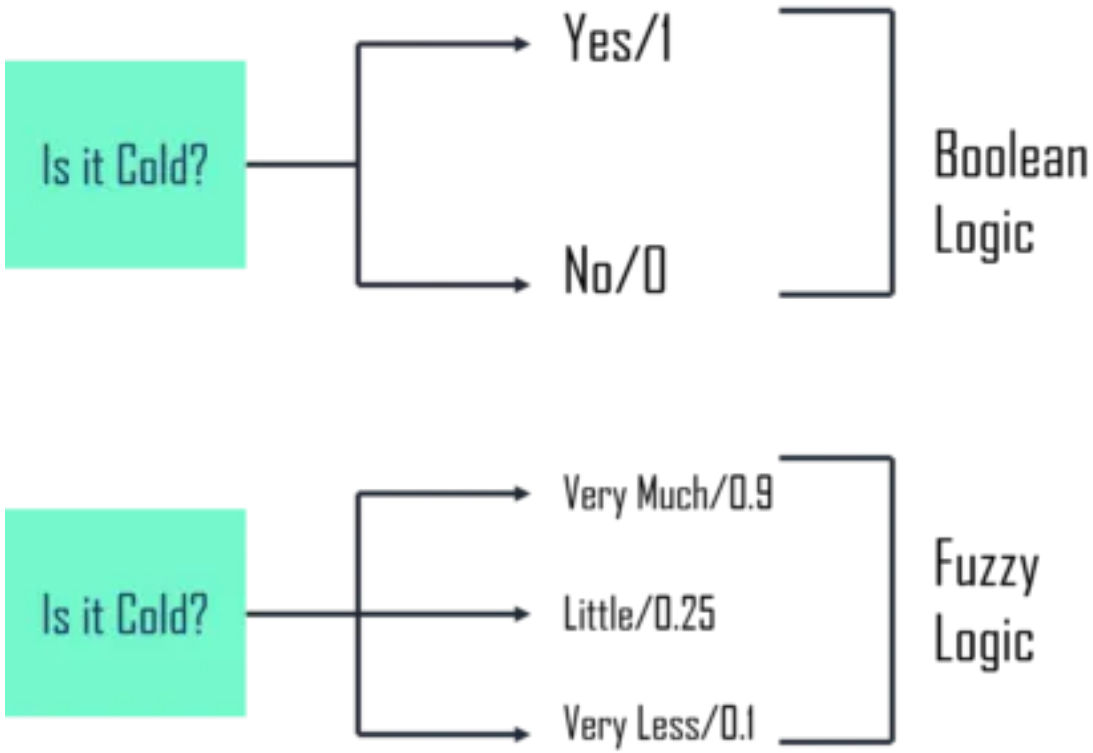

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Components of a Fuzzy Logic System

- **Fuzzification:**
  Converts crisp input (numerical) values into fuzzy sets.
  Example: Temperature = 35°C → “Hot” (0.8), “Warm” (0.2)
- **Knowledge Base:**
  Contains rules like:
  - IF Temperature is Hot THEN Fan Speed is High
  - IF Temperature is Warm THEN Fan Speed is Medium
- **Inference Engine:**
  Applies fuzzy reasoning to the rules to find which ones are true.
- **Defuzzification:**
  Converts fuzzy results back into a crisp output value.

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A **membership function (μ)** defines **how each crisp input value** (like temperature = 30°C) **belongs** to a **fuzzy set** (like *Cold*, *Warm*, *Hot*).
It assigns a **degree of membership** between **0 and 1**.

- **$\mu = 0$** → completely *not in* the set
- **$\mu = 1$** → completely *in* the set
- **$0 < \mu < 1$** → *partially* in the set

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AI for Knowledge Representation

- **Knowledge Representation (KR)** in Artificial Intelligence (AI) is the process of **storing information about the world** in a form that a computer system can use to **reason, understand, and solve problems intelligently** — similar to how humans use knowledge.
- Knowledge Representation (KR) in AI refers to **encoding information about the world into formats that AI systems can utilize to solve complex tasks.**
- This process enables machines to reason, learn, and make decisions by structuring data in a way that mirrors human understanding.
- It connects **human thinking** (concepts, facts, logic) with **machine reasoning** (data structures, algorithms).

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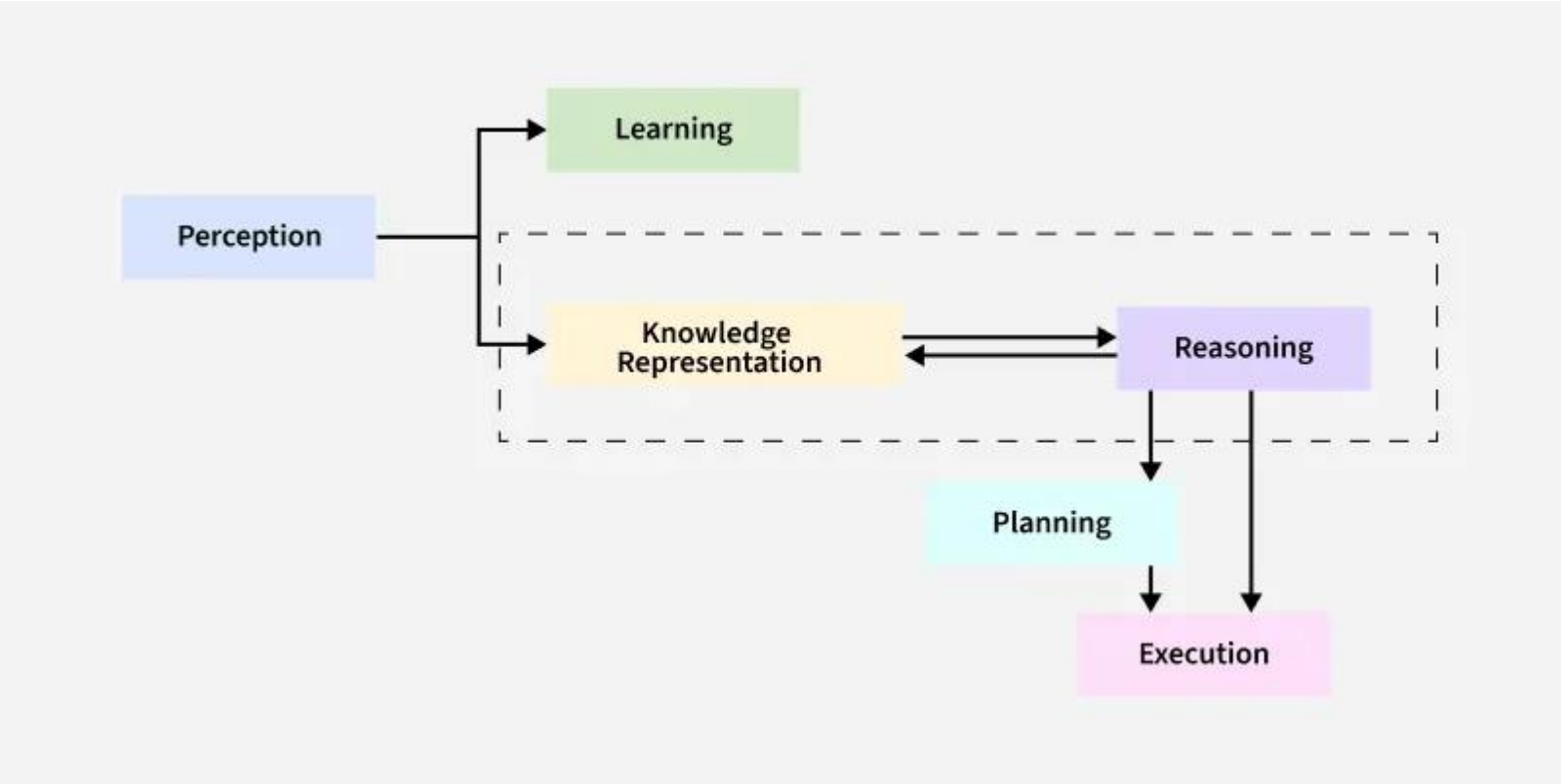

- Artificial intelligence systems operate on data. However, raw data alone does not lead to intelligence. AI must transform data into structured knowledge.
- KR achieves this by defining formats and methods for organizing information.
- With clear representations, AI systems solve problems, make decisions, and learn from new experiences.

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- **Knowledge as a Foundation**: Knowledge provides facts, rules, and data (e.g., traffic laws for self-driving cars). Without it, intelligence lacks the raw material to act.
- **Intelligence as Application**: Intelligence applies knowledge to solve problems (e.g., a robot using physics principles to navigate terrain).
- **Interdependence**: Static knowledge becomes obsolete without adaptive intelligence. Conversely, intelligence without knowledge cannot reason or learn (e.g., an AI with no medical database cannot diagnose diseases).
- **Synergy**: Effective AI systems merge robust knowledge bases (the *what*) with reasoning algorithms (the *how*). For example, ChatGPT combines vast language data (knowledge) with transformer models (intelligence) to generate coherent text.

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Core Methods of Knowledge Representation

Logic-Based Systems

Logic-based methods use formal rules to model knowledge. These systems prioritize precision and are ideal for deterministic environments.

Propositional Logic

Represents knowledge as declarative statements (propositions) linked by logical operators like AND, OR, and NOT. For example, "If it rains (A) AND the ground is wet (B), THEN the road is slippery (C)." While it struggles with complex relationships. Often follow the format "IF condition THEN conclusion." For instance, in a knowledge-based system, you might have:

- *IF an object is red AND round, THEN the object might be an apple.*

First-Order Logic (FOL)

Extends propositional logic by introducing variables, quantifiers, and predicates. FOL can express statements like, “All humans (∀x) are mortal (Mortal(x)).” It supports nuanced reasoning but demands significant computational resources.

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- **Structured Representations**
- **Semantic Networks**
  Represent knowledge as nodes (concepts) and edges (relationships). For example, "Dog" links to "Animal" via an "Is-A" connection. They simplify inheritance reasoning but lack formal semantics.
- **Frames**
  Group related attributes into structured "frames." A "Vehicle" frame may include slots like wheels, engine type, and fuel. Frames excel in default reasoning but struggle with exceptions.
- **Ontologies**
  Define concepts, hierarchies, and relationships within a domain using standards like OWL (Web Ontology Language). Ontologies power semantic search engines and healthcare diagnostics by standardizing terminology.

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Probabilistic Models

These systems handle uncertainty by assigning probabilities to outcomes.

- **Bayesian Networks**
  Use directed graphs to model causal relationships. Each node represents a variable, and edges denote conditional dependencies. For instance, a Bayesian network can predict the likelihood of equipment failure based on maintenance history and usage.
- **Markov Decision Processes (MDPs)**
  Model sequential decision-making in dynamic environments. MDPs help robotics systems navigate obstacles by evaluating potential actions and rewards.

Distributed Representations

Modern AI leverages neural networks to encode knowledge as numerical vectors, capturing latent patterns in data.

- **Embeddings**
  Convert words, images, or entities into dense vectors. Word embeddings like Word2Vec map synonyms to nearby vectors, enabling semantic analysis.
- **Knowledge Graphs**
  Combine graph structures with embeddings to represent entities (e.g., people, places) and their relationships. Google's Knowledge Graph enhances search results by linking related concepts.

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Declarative Knowledge ("Know-What")

- **Definition:**
  Declarative knowledge represents **facts and information** about the world — *what is true*.
  It describes **things, relationships, and properties**, but **not how** to use them.
- **Example:**
- “Paris is the capital of France.”
- “If it rains, the ground becomes wet.”

Used in:

- **Expert systems** (storing facts and rules)
- **Knowledge bases**
- **Logic programming (Prolog)**

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Procedural Knowledge ("Know-How")

- **Definition:**
  Procedural knowledge represents the **steps, procedures, or rules** for how to perform a task — *how to do something*.
  It focuses on **methods and processes** rather than facts.
- This type of knowledge is often stored in structured formats like databases, ontologies, and knowledge graphs.

Example:

- "To diagnose a disease, check symptoms, apply rules, and suggest treatment."
- *For instance, knowing how to solve a quadratic equation or how to drive a car falls under procedural knowledge. AI systems, such as expert systems and robotics, utilize procedural knowledge to execute tasks that require sequences of actions. This type of knowledge is often encoded in rule-based systems, decision trees, and machine learning models.*

Used in:

- **Rule-based systems**
- **Production systems**
- **Algorithmic processes (like searching, planning, reasoning)**

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Aspect	Declarative Knowledge	Procedural Knowledge
Meaning	“Know-what” (facts)	“Know-how” (process)
Representation	Facts, logic, relationships	Rules, procedures, algorithms
Example	“Fever is a symptom of flu.”	“If fever → test for flu.”
Flexibility	Easy to update or explain	Harder to extract or modify
Use in AI	Knowledge base	Inference or control mechanism

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- The objective of search in **Artificial Intelligence (AI)** is to find the path to solve different problems.
- The search in AI can be executed in two ways namely, **Forward Reasoning** and **Backward Reasoning**.

Forward Reasoning

- **Forward reasoning** is a process in artificial intelligence that finds all the possible solutions of a problem based on the initial data and facts.
- Thus, the forward reasoning is a data-driven task as it begins with new data.
- The main objective of the forward reasoning in AI is to find a conclusion that would follow.
- It uses an opportunistic type of approach
- In forward reasoning, the first step is that the system is given one or more constraints.
- The rules are then searched for in the knowledge base for every constraint.
- The rule that fulfils the condition is selected.
- Also, every rule can generate a new condition from the conclusion which is obtained from the invoked one.
- This new conditions can be added and are processed again.

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- **Backward reasoning** is the reverse process of the forward reasoning in which a goal or hypothesis is selected and it is analyzed to find the initial data, facts, and rules.
- Therefore, the backward reasoning is a goal driven task as it begins with conclusions or goals that are uncertain.
- The main objective of the backward reasoning is to find the facts that support the conclusions.
- Backward reasoning uses a conservative type of approach and flows from consequence to the incipient.
- The system helps to choose a goal state and reasons in a backward direction.
- The first step in the backward reasoning is that the goal state and rules are selected.
- Then, sub-goals are made from the selected rule, which need to be satisfied for the goal state to be true.

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Source

m({R})

m({N})

m({R, N})

Sensor 1

0.7

0.1

0.2

Sensor 2

0.6

0.3

0.1

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